

BHARATH P

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SUMMARY

Software Engineer with 1 year of experience building production AI applications and intelligent automation solutions. Developed LLM-powered systems using LangGraph, LangChain, and Azure OpenAI to automate analytics and conversational data exploration, reducing manual effort and improving decision-making. Experience in integrating AI with backend services and scalable data pipelines.

WORK EXPERIENCE

Junior Software Engineer at Peninsular Research Operation

July 2025 – Present

- Developed a LangGraph-based analytics workflow that automatically generated multi-level drill-down charts from natural language queries, enabling chart creation in under 3 seconds and allowing business users to explore enterprise datasets through hierarchical visualizations.
- Built an LLM-powered visualization agent using LangChain, Azure OpenAI, LIDA, and Plotly that transformed natural language requests into interactive charts, enabling self-service data exploration through a conversational interface.
- Developed a social media analytics pipeline using HDBSCAN, UMAP, and Azure OpenAI to cluster 5,000+ tweets daily into related discussions and generate automated insights, eliminating manual trend analysis.
- Built a large-scale OCR and document-extraction pipeline using Azure Computer Vision and Tesseract to process 2M+ scanned PDF document images, combining OpenCV based preprocessing with regex-driven structured-field extraction and parallelized multiprocessing to convert unstructured scans into clean, structured datasets.
- Designed and implemented a dual-mode authentication architecture using Keycloak, JWT validation, Angular route guards, and automatic token refresh, enabling secure and seamless user authentication across internal and external identity providers.
- Extended a Shopify loyalty platform with WebEngage integrations and Angular-based configuration tools, enabling clients to automate event-driven email campaigns without developer involvement.

Intern at Peninsular Research Operation

Jan 2025 – June 2025

- Developed background activity-tracking features using pedometer and geolocation data for a Flutter carbon footprint application, enabling continuous carbon footprint monitoring and improving user engagement.
- Developed a full-stack e-commerce platform using Angular, Node.js, Express.js, C#, MongoDB, and MySQL, delivering end-to-end functionality across product management, and order processing workflows.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript

AI & Machine Learning: LangGraph, LangChain, Azure OpenAI, RAG, Text Embeddings, Prompt Engineering, HDBSCAN, UMAP, OCR

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs

Databases: MongoDB, Pinecone, SQL, Redis

Tools & Cloud: Docker, Azure Functions, Git, GitHub, Azure DevOps

Frontend: Angular, React, Flutter

EDUCATION

Bachelor of Engineering in Computer Science Engineering CGPA: 8.15
Sathyabama University

2021 – 2025

PUBLICATION

Published Campus Commute: College Bus Tracking System in the 7th International Conference, IEEE Xplore.
<https://ieeexplore.ieee.org/document/11076377>